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# Bringing justice to informal adaptation to heat stress

Zakia Sultana, Heleen L. P. Mees, Bishawjit Mallick, Ilan Kelman, J. Magnolia Gilligan, Brianna Castro, Shanawez Hossain, Peter P. J. Driessen & Ajay Bailey



Informal adaptation can augment or replace formal adaptation with flexible and responsive actions, yet it can also reproduce and intensify existing injustices. We propose justice-oriented co-creation between informal and formal actions as a pathway for effective heat adaptation.

Adaptation practices emerge through interactions between formal systems and informal practices, rather than as a strict dichotomy[1,2]. Formal heat stress adaptation is planned and delivered through official institutions, policies and regulated service systems, such as heat action plans, public cooling centres, labour regulations and infrastructure investments[3,4]. By contrast, informal adaptation refers to self-organized responses undertaken by individuals, households and communities outside, alongside or in the absence of formal adaptation systems[1,5]. These responses emerge through everyday practices, social networks and locally negotiated arrangements rather than through official planning processes. For instance, reinforcing roofs with low-cost shading or insulation materials (for example, stretch fabric or plastic sheets), resting during the hottest hours, organizing water sharing at community taps, sharing heat-related information through community ties, and using courtyards, community centres and schools as temporary cooling spaces[5–7]. These informal adaptation practices can provide rapid and context-specific solutions to heat stress. Rooted in local knowledge, trust, reciprocity and close proximity to the most affected people[8], these practices often extend beyond the reach of formal adaptation programmes.

However, reliance on informal measures embodies deeper injustices[5,9]. Informal adaptation often relies on household resources, social networks and community support. As a result, its benefits are unevenly distributed across populations[6]. This inequity is particularly severe for disadvantaged groups who are insufficiently covered by formal adaptation and have to rely heavily on informal approaches. Informal adaptation, therefore, functions simultaneously as a means of coping with heat stress and as a symptom of systemic inequity that produces unequal and inequitable adaptation outcomes.

Informal adaptation should not be understood simply as a governance failure or as a substitute for formal adaptation. Rather, it is shaped through the selective withdrawal, rigidity and adaptation of formal systems[2,10]. Eliminating informal adaptation would erase many people's abilities to respond to marginalization, further amplified by climate change impacts; but the apparent success of informal adaptation may mask underlying vulnerabilities and inequities, and thus contribute to their persistence by removing them from view. Addressing this challenge requires greater attention to the ways in which formal and informal adaptation systems interact, support and shape one another. Here we propose a justice-oriented co-creation framework for reconfiguring interactions between formal and informal adaptation towards more just outcomes.

## Injustice mechanisms of informal adaptation

Informal adaptation is well grounded in local realities and frequently operates without sustained institutional support or enabling policy environments[1,2]. Thus, it may reproduce inequities by privileging those with greater resources, social connections or adaptive capacities. These limitations manifest through three recurring patterns and mechanisms of injustice: lack of recognition[1,2,5], regressive redistribution[1], and gaps in representation and accountability[1,8].

First, although informal adaptation practices often provide critical protection against climate change risks, their position outside formal planning frameworks means they are frequently treated as short-term coping mechanisms and denied resources that could allow them to provide deliberate and strategic resilience-building. This creates recognitional injustice by undervaluing the lived expertise and collective labour underpinning informal adaptation, and excluding informal actors from adaptation planning and decision-making processes[1,2,5]. As a result, an emphasis on informal adaptation can leave disadvantaged populations bearing primary responsibility for adaptation but excluded from power and access to resources because they lack voice and influence in formal governance processes.

Second, informal adaptation can result in regressive redistribution by shifting the economic and social costs of adaptation onto those with the least capacity, particularly where formal systems fail to provide adequate and equitable support. In informal markets, adaptation becomes conditional on purchasing power or labour flexibility: those with greater resources can secure private cooling or water access, while poorer tenants and street vendors remain exposed and must continue working under hazardous conditions[6,8]. In this way, adaptation burdens are redistributed downward socially and economically, reinforcing precarity, unequal exposure to heat risks and uneven access to urban resilience.

Third, informal adaptation generates representation and accountability gaps that translate into procedural injustice. Because informal actors operate outside formal regulatory recognition, they are excluded from decision-making processes and lack avenues to hold institutions accountable for adaptation outcomes or inaction. Legal ambiguity also enables intermediaries, including corrupt officials and criminal networks, to coerce and extract rents. As a result, adaptation decisions, including cooling-centre placement, relocation schemes and infrastructure design, are often made without meaningful participation from those most exposed. When informal practices conflict

## BOX 1

# Examples of hybrid heat adaptation practices

**Dhaka, Bangladesh**
Slum residents adapt to recurring heatwaves through self-directed practices such as increasing fluid intake, resting in shaded areas, running fans and relying on informal water-sharing within the community. These informal efforts provide immediate relief and help residents to cope with heat stress, but their effectiveness remains constrained by inadequate housing, unreliable electricity and limited formal access to potable water. Community-managed water provision, facilitated through collaborations between NGOs and community leaders, strengthen local coping practices during periods of extreme heat. This interaction reflects a form of hybrid governance in which formal and informal actors jointly shape adaptation outcomes. Yet, because access to water is mediated through local power relations, the benefits of these arrangements are not distributed evenly, leaving some residents better positioned to adapt than others[6].

**Cape Town, South Africa**
During the 2015–2017 drought, households across Cape Town adopted informal adaptation practices such as reducing water use, recycling water, purchasing bottled water, installing rainwater tanks and accessing alternative water sources. At the same time, formal adaptation measures included strict water restrictions, tariff increases, water metering and the Day Zero communication campaign[9]. Household adaptation responses were shaped by these municipal measures, while reduced household water consumption helped to support city-wide demand management during the crisis. These interactions revealed major inequalities and inequities in adaptive capacity, as wealthier households could secure alternative water supplies while poorer communities remained highly water-insecure. This demonstrates how formal water governance and everyday household adaptation co-evolve, while exposing uneven resilience outcomes across social groups[9].

**Nairobi, Kenya**
In the informal settlements of Kibera, Mathare and Mukuru, local temperatures and heat index levels regularly exceeded readings from the official Dagoretti meteorological station by several degrees Celsius, exposing residents to elevated heat-related health risks[12]. Informal settlements experienced intensified heat due to dense metal-roof housing, limited vegetation, poor ventilation and inadequate infrastructure. The formal city-scale monitoring systems underestimated actual exposure conditions within informal settlements, meaning that heat alerts based only on official station data could fail to protect vulnerable communities. Integrating settlement-specific data monitoring with formal meteorological and public health systems can support more targeted, equitable and locally responsive urban heat adaptation interventions[12].

**Lima, Peru**
In the informal settlement of José Carlos Mariátegui, residents (informal efforts including community-led greening and park creation) with support from an NGO (formal institutional support for park development) co-developed a small community park to improve environmental conditions and reduce heat stress[7]. Although the park produced only modest reductions in average air temperature, shaded areas substantially decreased heat stress exposure during peak daytime periods, lowering the occurrence of heat stress from more than 60% of daytime hours to below 20% under shaded conditions[7]. This collaboration demonstrates how community-led greening initiatives in informal settlements can complement formal urban heat adaptation planning by providing locally appropriate, low-cost interventions for low-income communities.

**Bhubaneswar, India**
Slum communities in Bhubaneswar engage with municipal authorities through both conflict and collaboration in the implementation of Smart City and climate change adaptation projects[10]. Informal actors, including community leaders, activists and community groups, negotiate land rights, housing and relocation processes with formal city agencies. Communities also organize self-help groups and local financial support systems. Formal institutions, including redevelopment schemes and climate change adaptation plans, interact with these informal practices through negotiation, selective collaboration and sometimes confrontation. This interaction demonstrates an evolving form of hybrid governance, where formal planning processes and everyday community practices co-work, although conflicts over eviction and unequal power relations often limit the effectiveness and inclusiveness of adaptation efforts[10].

with official regulations, they are labelled illegal rather than treated as essential public functions[2,8]. These accountability gaps deepen political exclusion, weaken institutional trust and shift the burden of adaptation onto already marginalized populations.

Together, these interconnected injustices reinforce structural inequities by concentrating climate change risks, adaptation burdens and political exclusion among already marginalized populations, ultimately reproducing uneven and unjust patterns of climate change adaptation. This highlights the need for hybrid adaptation approaches that better connect formal institutions with existing informal adaptive practices.

## Hybrid adaptation approaches

Current widespread heat exposure demands adaptation strategies that combine urban planning, social policy and local resilience-building, underscoring the need for hybrid governance that links formal institutions with informal community-level adaptation[10,11]. The emerging hybrid practices (Box 1) illustrate how adaptation can become more inclusive, accountable and responsive to everyday realities. These examples show that hybrid governance involves diverse forms of engagement among municipal authorities, non-governmental organizations (NGOs), community leaders and slum communities, with interactions spanning water provision, housing and land rights,

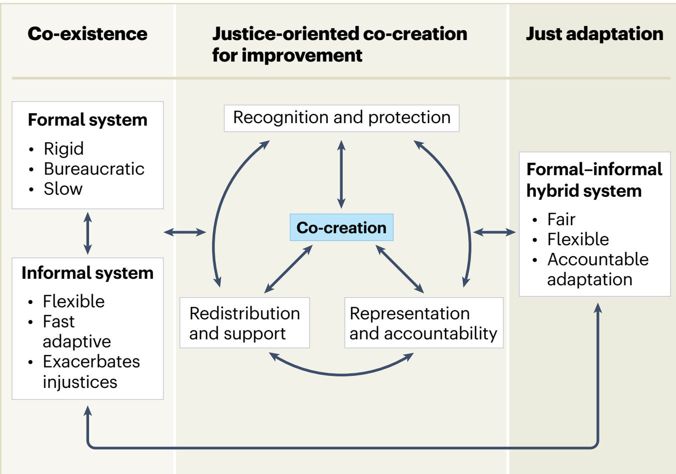


**Fig. 1 | Justice-oriented co-creation framework for just adaptation.** The left panel illustrates the existing formal and informal adaptation systems, which co-exist and may operate independently or interact with one another. Although these systems provide important adaptation responses, they can also reproduce or reinforce existing injustices. The central panel presents a justice-oriented co-creation process based on three institutional rearrangements: recognition and protection; redistribution and support; and representation and accountability. Together, these institutional rearrangements facilitate the development of a formal–informal hybrid adaptation system shown in the right panel. Because injustices may persist or emerge over time, the framework incorporates an iterative feedback loop to enable continuous learning, adjustment and improvement towards just adaptation.

urban planning, health monitoring, and community greening. These collaborations can provide effective and context-specific responses to heat stress, including more accurate exposure monitoring and immediate coping and cooling measures. Yet injustices persist, including the unequal distribution of benefits, conflicts and unequal power relations.

## Towards justice-oriented heat adaptation

Placing justice at the centre of heat stress adaptation requires attention not only to how heat risks and resources are distributed, but also to whose knowledge, priorities, power and adaptive practices shape adaptation decisions[8]. This dual concern highlights that adaptation is not only a technical challenge but also a fundamentally political one, in which choices about protection, investment and inclusion have uneven consequences across social groups. Current governance approaches to heat, whether driven by planned adaptation, private-sector mechanisms or community initiatives, each contribute to building resilience, yet none adequately address the structural inequities associated with informal adaptation[5,8]. Together, these limitations point to the need for an approach that meaningfully bridges formal and informal adaptation practices while directly addressing the injustices that shape vulnerability. It requires shifting adaptation from compensatory responses to address the risks of informal adaptation towards transforming the conditions that reproduce injustice.

To this end, we propose a justice-oriented co-creation framework (Fig. 1). Rather than treating formal and informal systems as parallel spheres that merely co-exist, co-creation reframes adaptation as a shared governance process in which authority, resources and recognition are more deliberately rebalanced between institutions and actors already adapting in place[1,2,10]. This reframing shifts attention from whether adaptation occurs to how it is governed and whose capacities are strengthened through that process. In this framing, the effectiveness of adaptation is measured not only by the delivery of physical infrastructures but also by the equitable flow of resources to those most exposed, the strengthening of local adaptive capacities and the dignity with which adaptation unfolds.

Justice-oriented co-creation differs from conventional consultation or stakeholder participation because it requires shared authority over the design, implementation and evaluation of adaptation interventions. Its purpose is not only to include informal actors in adaptation processes but also to rebalance decision-making authority between institutions and communities engaged in informal adaptation. Through this process, communities can help shape adaptation priorities, implementation and evaluation over time. Such shared decision-making authority is important because participation without influence can reproduce existing power imbalances rather than reduce them.

The proposed framework is structured around three institutional rearrangements that address the interconnected injustices of lack of recognition, regressive redistribution, and gaps in representation and accountability (see Fig. 1 for conceptual framework and Box 2 for illustrative examples). First, recognition and protection involve recognizing and acknowledging the legitimacy of informal knowledge, practices, values and actors, and integrating them into adaptation planning and governance. Second, redistribution involves directing financial, technical and infrastructural support to disadvantaged populations to reduce unequal and inequitable burdens of adaptation and strengthen adaptive capacity. Third, representation and accountability involve creating shared decision-making arrangements through which informal actors have the power to influence adaptation priorities and hold institutions accountable for outcomes. Together, recognition establishes legitimacy, redistribution provides resources and support, and representation enables influence over decisions and accountability for outcomes. These shifts move authority over knowledge, resources and decision-making beyond formal institutions and towards more inclusive adaptation processes.

Co-creation processes must also remain grounded in the social and spatial relations through which informal adaptation already operates, rather than being treated as stand-alone institutional platforms. Public institutions can support these processes through planning mechanisms, funding arrangements, regulatory flexibility and long-term institutional support, while community organizations and local networks contribute context-specific knowledge and adaptive practices[3,5,10]. At the same time, without explicit attention to power relations, co-creation risks reproducing existing hierarchies rather than transforming them. Such co-creation arrangements can strengthen institutional responsiveness while preserving the flexibility and proximity that often make informal adaptation effective.

**Zakia Sultana** [1,2,3] ✉, **Heleen L. P. Mees** [1], **Bishawjit Mallick** [2], **Ilan Kelman** [4], **J. Magnolia Gilligan** [5], **Brianna Castro** [6], **Shanawez Hossain** [7], **Peter P. J. Driessen** [1] **& Ajay Bailey**[2]

[1]Copernicus Institute of Sustainable Development, Faculty of Geosciences, Utrecht University, Utrecht, The Netherlands. [2]Department of Human Geography and Spatial Planning, Faculty of Geosciences, Utrecht University, Utrecht, The Netherlands. [3]Department of Environmental Science and Disaster Management, Gopalganj Science and Technology University, Gopalganj, Bangladesh. [4]Institute for Risk and Disaster Reduction, University College London (UCL), London, UK. [5]Department of Earth & Environmental Sciences, Vanderbilt University, Nashville, TN, USA.

## BOX 2

# Illustrative examples of rearrangements for just heat stress adaptation

**Recognition and protection**

Communities facing extreme heat often hold critical, localized knowledge and everyday practices about cooling, shading and ventilation, yet this expertise is rarely valued in formal adaptation planning. Recognition requires moving beyond treating informal adaptation as short-term coping mechanisms and validating community organizations, settlement committees and cooperatives as legitimate and knowledgeable adaptation actors within planning processes. Crucially, recognition must be tied to protection, ensuring that when formal improvements are made, informal actors are protected from eviction or resource exclusion. This requires not only symbolic inclusion, but also institutional protection and meaningful incorporation of lived experience into adaptation planning. Such recognition can improve the relevance and effectiveness of heat stress adaptation interventions. For example, Ahmedabad's Heat Action Plan operationalized recognition by integrating community outreach and local vulnerability knowledge through door-to-door campaigns, community-based warning dissemination, and verbal heat alerts targeting high-risk populations such as outdoor labourers and informal workers[13].

**Redistribution and support**

Heat places disproportionate burdens on low-income households, which must finance cooling materials, shade or additional water despite having the fewest resources. Redistribution can reduce these inequities by directing financial and infrastructural support towards highly exposed communities. For instance, in Kuwait and across the Arabian Peninsula, low-income populations experience disproportionate heat exposure due to poorly insulated housing, limited cooling access, and unequal and inequitable healthcare provision[14]. Investments in passive cooling interventions can reduce urban heat exposure and provide accessible heat-relief benefits when equitably distributed across neighbourhoods. Vegetation-based shading and green–blue infrastructure, including street trees, parks, wetlands and green walls, can improve thermal comfort and contribute to local cooling, particularly when investments are prioritized in highly exposed and underserved communities[4,7]. Community-accessible funding, municipal grants and neighbourhood upgrading initiatives can further support locally appropriate adaptation[8]. In this way, adaptation systems shift from expecting households to absorb heat risks on their own to ensuring that the resources required for cooling and safety are shared fairly.

**Representation and accountability**

Representation moves beyond acknowledging who informal actors are to structuring how much power they hold over resources and policy. It addresses the reality that participation without influence simply reproduces existing hierarchies. Communities experiencing extreme heat need sustained authority over adaptation decisions rather than one-off, symbolic consultations. This requires shared governance arrangements where local groups, city agencies and service providers co-design and evaluate interventions[15]. For example, neighbourhood-scale co-production initiatives in Phoenix (USA) involved residents and community organizations in designing hyper-local heat action plans through iterative workshops, operationalizing sustained influence beyond a one-off consultation[4]. Similarly, the Middle East and North Africa region's growing urban populations face compounded environmental and political risks, prompting a shift towards resilience approaches that integrate community-based heat stress adaptation, social inclusion and strengthened local governance[15]. Building-focused interventions such as cool-roof retrofits in informal settlements have shown substantial reductions in indoor heat stress, highlighting the importance of including informal urban areas within formal adaptation planning. These arrangements strengthen accountability within formal institutions while also making the decision-makers within those institutions answerable for their choices, rather than displacing responsibility onto individuals and communities with the least power[6]. Representation is therefore not an end in itself, but a way to align formal planning with lived experience and prevent heat interventions from reinforcing existing inequities.

[6]Yale School of the Environment, Yale University, New Haven, CT, USA. [7]Department of Global Studies and Governance, Independent University, Bangladesh (IUB), Dhaka, Bangladesh.
e-mail: z.sultana@uu.nl

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### Competing interests

The authors declare no competing interests.